PAGE No.
DATE / /

Case Study 1

Bright Mart is a medium size online shopping company customers use its website & mobile application to purchase products.

The company stores :-

i) The customer names
ii) Email address
iii) Phone no.
iv) Delivery address
v) Password.
vi) order info
vii) Payment info

Bright Mart has around 100 employees
The company recently had new employee give employe access to different internal system.

Howeves, the company has the foldowing security weakness

- Some employess use simple password such as Bright Mart 1234.

- Many employees use same password for multiple system.
- Software updates are sometimes delayed
- The company does not regularly conduct employees cybersecurity training.
- Employees can access sensitive info that they do not necessarily need for their jobs.
- Backups are performed only once a month.

The incident

One monday morning, an employee named Rahul receives an email, subject : Urgent Your Account
The email appears to be come from company IT depot
It says
Your company account will be suspended today

Rahul clicks the link

He enters his username & password.

PAGE No.
DATE / /

Stage 1 - Unauthorised Access

The attacker uses Rahul's credentials to access system. Because Rahul uses the same passc for another system, the attacker is also a to access that system. The attacker discovers that Rahul has access to customer information.

Stage 2 - Malware Attack

The attacker sends another email to Rahul cont a malacious attack. Rahul opens the attach Malware is installed. The malware begins communicating with the attacker system The attacker now attempts to move from Rahul computer to other systems within Bright Mart's n

stage 3 - Ransomware

Two days later, employess arrive at work & discover that they cannot access importen files. The company ... customer

PAGE No.
DATE / /

Case study 1 - Solution

Q) What cybersecurity problem has occurred?

- Phishing
- Credential theft
- Unauthorized access
- Data breach

Q) Which vulnerability allowed the attackes to move further into the organization?

- Weak password
- Excessive user priviliges
- Lack of MFA
- Poor network security
- Lack of employees
- Outdated softwase.

Q) What threats are present in this case

- Phishing
- Malware
- Ransomware
- Credential theft
- Unauthorized access

PAGE No.
DATE / /

q) What weakness in Bright Mart's security allowed the attack to succeed?

Vulnerability	How it helped the attacker
Weak password	Easier to compromise accounts
Password reuse	one stolen password gives access to system
No MFA	stolen password may be enough to login
Lack of training	Employees falls for phishing
Delayed Updates	Vulnerable software may be exploited.
Excessive Privilages	Attacker can access more data
Infrequent backups	Difficult to recover after ransomware

Q) Was confidential info exposed?

Yes, customer info was accessed / downloaded by attack

Q) Was info changed or damaged?

Yes, the attacker could modify or encrypt files.

Q) Were system ~~able to~~ available to legitimate users?

No, the website & internal system became unavailable because of ransomware.

So:

Confidentiality → Affected
Integrity → Affected
Availability → Affected

Group - 4

Q) What should Bright Mart do to prevent this from happening again?

- Use strong password
- Implement MFA
- Provide regular cyber security awareness training
- Conduct phishing awareness training exercise.
- Keep software updated
- Follow the principle of least privilege
- segment important networks.
- Maintain regular backups
- Encrypt sensitive data
- Deploy antivirus / endpoint activity
- Monitor suspicious activity
- Develop an incident - response plan

Q) Who is responsible for cyber security?

Not only IT department. Every employee who uses the organisation system has a role in it.

Q) Was phishing the only problem?

No, Phishing was the initial attack, but weakness such as password reuse, lack of MFA.

Q) What was the vulnerability?

Lack of employee awareness. was a major vulnerability because the attack started when Rahul clicked the link.

PAGE No.
DATE 14/1

Quick Assessment

Q1) What was the initial attack against BrightMart?
→ Phishing & credential theft

Q2) Give three vulnerabilities the helped the attacker
→ No MFA, weak password & lack of awareness.

Q3) Which three security principles of the CIA Triad were affected?
→ Confidentiality, integrity & availability were affected.

Q4) What was the impact of ransomware on BrightMart
→ The ransomware encrypted files & caused BrightMart website & internal system to become unavailable.

Q5) Suggest three security controls Bright mart should implement?

→ 1) implement of MFA
2) Maintain regular backup
3) Keep software updated.

Seema
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